GAZE: Grounding-Aware Zone-Adaptive Visual Token Selection with Focal–Spatial Synergy for Efficient MLLMs

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Abstract

Visual token compression is essential for improving the inference efficiency of multimodal large language models (MLLMs). However, existing training-free methods suffer from attention drift and degrade significantly at low retention rates. Training-based methods lack semantic guidance and incur high training costs, often requiring large-scale data to achieve generalization. To address these issues, we propose GAZE, a grounding-aware end-to-end token pruning framework. Specifically, GAZE injects foreground–background priors into a lightweight scorer via a localization-aware ranking loss, while employing dynamic budget allocation to enhance generalization. Furthermore, GAZE features a Focal–Spatial dual-channel selection mechanism: the Focal channel focuses on semantically salient tokens, whereas the Spatial channel ensures structural coverage via grid-based priors. Both channels share a redundancy penalty to balance discriminativeness and diversity. Experiments show that GAZE requires only 70 minutes of training on a single RTX 4090, yet delivers competitive or superior performance across multiple benchmarks. Notably, retaining only 30% of visual tokens, GAZE preserves 98% of full-token baseline accuracy on average, and achieves a 1.8× prefill speedup at 10% retention.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Neural networks**.

Keywords

Visual Token Pruning, Multimodal Large Language Models, Visual Grounding, Token Compression, Focal–Spatial Selection

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1 Introduction

In recent years, multimodal large language models (MLLMs) [2, 23] have achieved remarkable progress in complex tasks such as visual question answering, document understanding, and visual reasoning. However, due to the quadratic computational complexity of the self-attention mechanism in Transformers [35], the computational cost and memory overhead scale dramatically when processing high-resolution images or long videos. Meanwhile, extensive empirical evidence suggests [6, 31] that visual tokens exhibit substantial redundancy, and retaining only a small subset of critical tokens suffices for MLLMs to accomplish the majority of vision-language tasks. Visual token pruning has thus emerged as a promising research direction, aiming to significantly reduce the number of visual tokens with minimal accuracy degradation, thereby enabling more efficient and scalable multimodal inference.

Existing pruning methods can be broadly categorized into two groups. Training-free methods [1, 4, 6, 10, 31, 38] rely on heuristic criteria derived from attention scores or token similarity, but are fundamentally constrained by the internal feature distributions of specific models, and susceptible to biases such as attention artifacts [12, 17, 36] and feature distribution skew [9], leading to

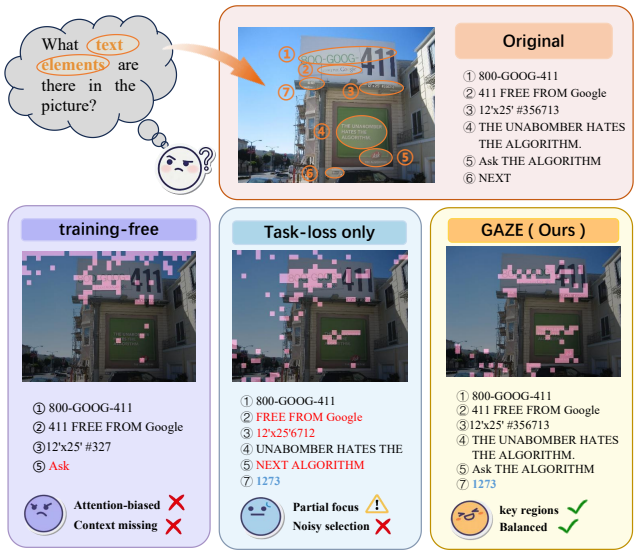


Figure 1: Illustration of different visual token pruning strategies (10% retention, Qwen2.5-VL-7B). Erroneous outputs due to pruning are shown in red; correct outputs absent in the full-token baseline but recovered after pruning are shown in blue. Training-free method (FastV) suffers from attention bias and misses peripheral text. Task-loss-only learning reduces bias but still produces hallucinations and misreadings. GAZE achieves more balanced spatial coverage and substantially more accurate outputs.

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sharp performance degradation under aggressive compression ratios. Learning-based methods [13, 42] train learnable auxiliary predictors to guide token selection, maintaining more stable accuracy even under more aggressive compression ratios, yet still suffering from two notable limitations. First, the training objective lacks explicit guidance on the semantic importance of tokens; the scorer can only implicitly discover visual saliency through indirect signals from the downstream task loss. Second, the Top-K selection strategy at inference simply truncates tokens by score ranking without spatial diversity constraints, tending to concentrate retained tokens in a few high-response regions while neglecting contextual information from peripheral and background areas, which may limit performance in complex scenes containing multiple spatially dispersed objects. As illustrated in Fig. 1, GAZE effectively supplements contextual information from peripheral and background regions while preserving semantic discriminability, in contrast to the typical failure modes of both method categories.

The analysis reveals an inherent conflict in existing learning-based pruning: retaining tokens solely by importance ranking leads to local clustering, whereas enforcing uniform sampling sacrifices semantic discriminability. At the same time, research in visual grounding [16, 21, 29, 39] has demonstrated that bounding box annotations naturally encode rich foreground–background spatial semantic priors, and this ability to distinguish semantically important regions from information-sparse backgrounds can directly serve as supervisory signals for token importance estimation. Motivated by this insight, we propose **GAZE**, whose core idea is to introduce visual grounding annotations into token pruning training as explicit semantic priors to enhance an end-to-end learnable framework. Specifically, GAZE incorporates two complementary and mutually reinforcing improvements. First, we introduce grounding-aware supervision during training: a ranking loss constructed from bounding box annotations directly guides a lightweight Visual Importance Scorer (VIS) to assign higher scores to semantically salient regions, and the retention budget for each image is dynamically adjusted accordingly, enabling VIS to generalize well across diverse compression ratios. Second, at the token selection level, we design a Focal–Spatial dual-channel mechanism to replace the standard Top-K operation: the Focal channel focuses on the most discriminative core tokens, while the Spatial channel supplements global context through spatially uniform grid priors; both channels share a redundancy penalty that suppresses semantically overlapping candidates to promote diversity in the final selection. This mechanism is uniformly applied during both training and inference in an end-to-end differentiable manner.

Experiments show that GAZE consistently outperforms existing state-of-the-art methods across various compression budgets. Notably, retaining only 30% of visual tokens, GAZE preserves 98% of full-token baseline accuracy on average, and achieves a 1.8× prefill speedup at 10% retention. Across multiple image understanding benchmarks, GAZE exhibits consistent and significant advantages under varying compression budgets and backbone architectures.

Our main contributions are summarized as follows:

- We propose GAZE, which introduces visual grounding annotations as explicit supervisory signals into an end-to-end differentiable token selection framework. Through a bounding box ranking loss, GAZE provides VIS with foreground–background spatial semantic priors and dynamically adjusts the retention budget per image, enabling VIS to generalize well to arbitrary fixed compression ratios at inference time.
- We design a Focal–Spatial dual-channel selection mechanism to replace the standard Top-K operation, simultaneously ensuring semantic precision and spatial coverage through saliency-prioritized de-redundancy selection and grid-guided contextual supplementation.
- The proposed modules are fully decoupled from the backbone network, requiring no modification or retraining of the backbone, and are compatible with acceleration techniques such as FlashAttention [7, 8]. Extensive experiments on multiple benchmarks validate the effectiveness of GAZE across various compression budgets and model scales, comprehensively surpassing existing baseline methods.

2 Related Work

2.1 Visual Token Pruning

Visual token pruning aims to identify and remove redundant visual tokens so as to improve the inference efficiency of MLLMs while preserving model performance. Depending on whether additional training is required, existing approaches can be broadly categorized into **training-free methods** and **training-based methods**.

Training-free methods. Training-free methods leverage internal heuristic signals of the model to estimate token importance in a plug-and-play manner, and can be further organized along three main lines. Attention-based approaches such as FastV [6] determine token importance via the internal attention scores of the LLM and prune tokens at shallow layers; however, they are inherently susceptible to the attention-sink bias [12, 17, 36], leading to noticeable performance degradation at high compression ratios. Redundancy-based approaches such as LLaVA-PruMerge [31] cluster and merge tokens with similar features, and V^2Drop [4] takes a cross-layer variation perspective, measuring the L2 distance of each token's representation between adjacent LLM layers and progressively dropping tokens exhibiting minimal variation; however, the selection criteria of these methods lack an explicit distinction between foreground and background semantics. Diversity-based approaches such as DivPrune [1] employ farthest-point sampling to optimize token coverage, yet purely uniform sampling does not differentiate between semantically dense and sparse regions, failing to reflect task-relevant saliency; PruneSID [10] jointly preserves diverse tokens through semantic clustering and intra-group Non-Maximum Suppression, yet its selection does not leverage semantic feedback from downstream tasks to guide the filtering process. Overall, despite their respective merits, training-free methods uniformly rely on hand-crafted heuristics and lack both task-relevant semantic supervision and spatial awareness, fundamentally limiting their robustness under high compression ratios.

Training-based methods. Training-based methods introduce learnable components to guide token selection, trading moderate training overhead for improved pruning accuracy. For instance, Dynamic-LLaVA [13] trains a lightweight predictor with the multimodal downstream task loss to generate a numerical decision score

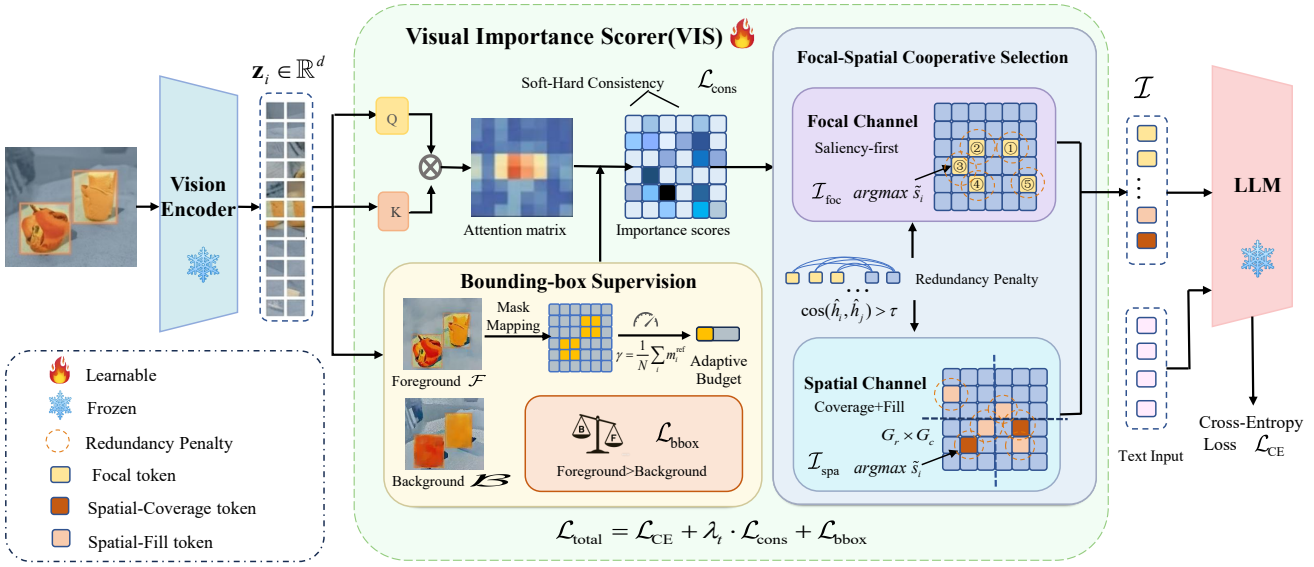


Figure 2: The overall pipeline of GAZE. A trainable Visual Importance Scorer is inserted between a frozen vision encoder and an LLM. It assigns token-level scores under bounding-box supervision, and selects the most informative tokens via Focal–Spatial cooperative selection with a shared redundancy penalty.

for every token, pruning low-scoring redundant tokens and retaining only high-scoring salient ones for subsequent computation. VisionSelector [42] further introduces differentiable Top-K selection coupled with a soft–hard mask consistency constraint and a curriculum annealing schedule that progressively aligns selection behavior between training and inference, achieving a more favorable compression-performance trade-off across multiple benchmarks. Although training-based methods demonstrate superior robustness over their training-free counterparts, existing work still shares the following common limitations: (1) the training objective lacks *explicit* guidance on token-level semantic importance: the scorer can only implicitly discover visual saliency through indirect gradient signals from the downstream task loss, without direct supervision that distinguishes foreground from background; (2) the Top-K selection strategy at inference lacks spatial diversity constraints, as ranking tokens purely by score tends to concentrate the retained set in a few locally high-response regions, neglecting peripheral and contextual information that is crucial for holistic visual understanding; (3) adaptivity is confined to the inference stage, as training typically adopts a fixed compression ratio, limiting the scorer's generalization to varying retention ratios at test time.

2.2 Spatial Visual Understanding

Visual grounding aims to localize target regions in an image given a natural language description, serving as a fundamental capability that bridges linguistic semantics and spatial locations. From the early end-to-end detection framework MDETR [16] to recent efforts that directly integrate grounding capabilities into multimodal large language models, such as Kosmos-2 [29], Shikra [5], and Ferret [39], this line of research has progressively advanced from text-box alignment to multi-granularity spatial understanding encompassing points, boxes, and free-form regions. Subsequent work including GroundingGPT [21] and Pink [37] further extends grounding capabilities into the broader multimodal instruction-tuning paradigm. These studies reveal that grounding annotations inherently encode rich foreground–background spatial-semantic priors. Such ability to partition an image into semantically salient regions and information-sparse backgrounds aligns closely with the core requirement of visual token importance estimation. Motivated by this observation, we introduce visual grounding annotations into the token pruning training pipeline, providing the scorer with semantic supervision, spatial distribution priors, and compression-ratio diversity in a unified framework.

3 Method

3.1 Overall Framework

Given a visual input and a textual instruction, our goal is to select the most informative subset from the visual token sequence and feed it into the LLM, substantially reducing computational cost while preserving task performance. GAZE inserts a lightweight **Visual Importance Scorer (VIS)** between the visual encoder and the language model to compute per-token importance score and retain k informative tokens. Figure 2 illustrates the overall pipeline. Specifically, GAZE introduces two complementary improvements. **Grounding-aware supervision with adaptive budget allocation.** A ranking loss constructed from bounding-box annotations provides the scorer with explicit foreground–background semantic supervision, while the retention budget is dynamically adjusted per image according to its foreground proportion. **Focal–Spatial dual-channel cooperative selection with shared redundancy penalty.** The Focal channel targets semantically salient core tokens,

the Spatial channel supplements spatial coverage via a grid prior, and the two channels share a redundancy penalty to jointly promote discriminability and diversity. The compressed token subset is concatenated with the text tokens and fed into the LLM for autoregressive generation; the entire model is optimized end-to-end without multi-stage training.

3.2 Bounding-Box-Guided Supervision and Adaptive Budget Allocation

A VIS trained solely with the downstream task loss lacks explicit spatial-semantic guidance and tends to assign high scores to textural artifacts in background regions that exhibit strong self-attention responses. To address this, we leverage bounding-box annotations to enhance scorer training from two complementary perspectives: constructing a localization-aware ranking loss that provides explicit semantic supervision, and dynamically adjusting the retention budget for each image accordingly.

Visual importance scoring. Given a visual input x_v and the corresponding textual input x_t, the visual encoder encodes the image into a sequence of continuous visual tokens $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^{N}$, $\mathbf{z}_i \in \mathbb{R}^d$, where N is the total number of tokens and d is the feature dimension. VIS projects the visual features into queries and keys and derives token-level scores via global correlations:

$$Q = \mathbf{Z}W_q, \quad K = \mathbf{Z}W_k, \quad A = \frac{QK^\top}{\sqrt{d_h}}, \quad s_i = \frac{1}{N}\sum_{j=1}^{N} A_{ij}, \tag{1}$$

where W_q, W_k are learnable parameters and d_h is the projection dimension. We denote the initial importance score vector as $\mathbf{s} = [s_1, \ldots, s_N]^\top \in \mathbb{R}^N$.

Localization-aware ranking loss. Using the bounding-box annotations from the Visual CoT dataset [32], we partition the visual tokens according to their spatial positions into a foreground set $\mathcal{F}$ and a background set $\mathcal{B}$, corresponding to tokens that fall inside any annotated box and those outside all annotated boxes, respectively. A ranking loss is then introduced to enforce a clear margin between their importance scores:

$$\mathcal{L}_{\text{bbox}} = \text{softplus}\big(-(\bar{s}_{\mathcal{F}} - \bar{s}_{\mathcal{B}} - \delta)\big), \tag{2}$$

where $\bar{s}_{\mathcal{F}} = \frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}} s_i$, $\bar{s}_{\mathcal{B}} = \frac{1}{|\mathcal{B}|}\sum_{j\in\mathcal{B}} s_j$, $\delta > 0$ is a margin hyperparameter that ensures sufficient discriminability in the score distribution, and $\text{softplus}(x) = \log(1 + e^x)$. The loss is minimized when all foreground scores exceed the background scores by at least δ, thereby ensuring a clear separation between the two groups.

Adaptive budget allocation. Images differ substantially in information density: scenes with dense foreground objects require more tokens to be retained, while images with simple backgrounds can tolerate more aggressive pruning. Existing methods train with a single fixed pruning ratio, causing the scorer to overfit to one specific compression rate and severely limiting its generalization. We address this with an adaptive budget allocation strategy that dynamically determines a per-image retention budget from the foreground coverage ratio, exposing the scorer to a wide spectrum of compression rates during training. This diversity effectively augments training over the compression budget, yielding robust performance at any user-specified inference ratio β'.

Concretely, we project the bounding-box annotations onto the visual token grid to obtain a binary reference mask $\mathbf{m}^{\text{ref}} \in \{0,1\}^N$, where $m_i^{\text{ref}} = 1$ indicates that token i falls within a foreground region. The foreground ratio $\gamma = \frac{1}{N}\sum_i m_i^{\text{ref}}$ captures the information density of each image and provides a principled basis for the per-image retention rate:

$$\beta = \text{clip}(\gamma, \rho_{\min}, \rho_{\max}), \qquad k = \lceil \beta N \rceil, \tag{3}$$

where $\rho_{\min}$ and $\rho_{\max}$ are lower and upper bounds on the retention ratio, preventing training instability from extreme compression. At inference time, no bounding-box annotations are needed. The scorer is applied directly with the user-specified retention ratio β'.

3.3 Focal–Spatial Cooperative Token Selection

Focal–Spatial cooperative selection. Existing strategies inevitably face a trade-off between saliency and coverage at high compression ratios: ranking tokens purely by importance score causes the retained set to cluster in locally high-response regions, losing global context; enforcing spatially uniform sampling, on the other hand, sacrifices semantic discriminability. To address this, we split the total retention budget k into two complementary channels: a Focal channel with budget k_{foc} that focuses on the most discriminative core tokens, and a Spatial channel with budget k_{spa} that supplements structural and contextual information. The two budgets satisfy:

$$k_{\text{foc}} + k_{\text{spa}} = k. \tag{4}$$

Both channels greedily select tokens based on the importance scores $\mathbf{s}$, yet ranking by scores alone tends to repeatedly retain semantically similar tokens within locally high-response regions, thereby wasting the limited budget on redundant information. We therefore introduce a shared redundancy penalty mechanism for both channels that effectively preserves salient tokens while promoting semantic complementarity among the selected set. We first describe this penalty mechanism and then detail the specific selection strategies for the Focal and Spatial channels.

Redundancy penalty. Ranking solely by importance scores tends to repeatedly retain semantically similar tokens in locally high-response regions. To mitigate this issue, we introduce a redundancy penalty at each step of the greedy selection, reducing the scores of candidates that are semantically highly overlapping with already-selected tokens.

Let the feature representation of the i-th visual token be $\mathbf{h}_i$, and its ℓ_2-normalized counterpart be $\hat{\mathbf{h}}_i = \mathbf{h}_i / \|\mathbf{h}_i\|_2$. Denoting the set of all currently selected tokens as $\mathcal{A}$, the redundancy penalty for a candidate token i is defined as:

$$p_i = \max\big(0, \max_{j\in\mathcal{A}} \cos(\hat{\mathbf{h}}_i, \hat{\mathbf{h}}_j) - \tau\big), \tag{5}$$

where τ is a similarity threshold. A non-zero penalty is incurred only when the maximum cosine similarity between the candidate and any already-selected token exceeds τ, thereby avoiding excessive suppression of normal local correlations. The final selection score for a candidate token is then defined as:

$$\tilde{s}_i = s_i - \alpha\, p_i, \tag{6}$$

where α is an adaptive penalty coefficient. To match the penalty magnitude to the score distribution of the current sample, we adopt

an interquartile-range (IQR)-based scale matching strategy:

$$\alpha = \begin{cases} \dfrac{\mathrm{IQR}(\{s_i\}_{i\in\Omega})}{\mathrm{IQR}(\{p_i\}_{i\in\Omega}) + \epsilon}, & \mathrm{IQR}(\{p_i\}_{i\in\Omega}) > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

where $\mathrm{IQR}(\cdot)$ denotes the interquartile range, Ω is the current candidate set, and ϵ is a small constant to prevent division by zero. This design avoids the scale mismatch that a fixed penalty coefficient would exhibit across different samples, enabling the redundancy constraint to adapt to the score distribution automatically.

Focal channel: saliency-first greedy selection. The Focal channel performs saliency-first greedy selection over the global candidate set, progressively building the core token index set $\mathcal{I}_{\text{foc}}$. Since the Focal channel is designed to execute prior to the Spatial channel, the selected set $\mathcal{A}$ is initially empty, and the token with the highest raw importance score is first chosen as a seed. At each subsequent step, the token with the highest composite score is selected from the remaining candidates:

$$i^* = \arg\max_{i\notin\mathcal{I}_{\text{foc}}} \tilde{s}_i, \tag{8}$$

and added to $\mathcal{I}_{\text{foc}}$, with the penalty updated accordingly, until exactly k_{foc} tokens have been selected. Throughout this process, high-saliency regions are consistently prioritized while suppressing multiple high-scoring tokens from concentrating on the same object or local texture, thereby jointly improving the compactness and effectiveness of the retained core visual information.

Spatial channel: coverage-oriented context supplementation. After Focal selection is complete, we perform coverage-oriented context supplementation on all remaining tokens not yet selected by the Focal channel, with remaining budget $k_{\text{spa}} = k - k_{\text{foc}}$. At this point, the selected set $\mathcal{A}$ in the penalty computation contains $\mathcal{I}_{\text{foc}}$ as well as the Spatial tokens selected so far, thereby simultaneously and effectively suppressing both intra-channel clustering and cross-channel redundancy.

Since selecting directly from the global residual candidates alone cannot reliably guarantee spatially uniform coverage, we introduce a structured grid prior. Let the spatial dimensions of these tokens on the merged visual grid be $H \times W$; the grid partition is adaptively constructed according to the budget k_{spa}:

$$G_c = \left\lceil \sqrt{k_{\text{spa}} \cdot \tfrac{W}{H}} \right\rceil, \quad G_r = \left\lceil \frac{k_{\text{spa}}}{G_c} \right\rceil, \tag{9}$$

yielding $G_r \times G_c$ grid cells, into which all candidate tokens are mapped according to their spatial positions. A two-stage selection then follows. **Stage 1** serves as the coverage stage: from each non-empty grid cell, the highest-scoring token is selected as a candidate, thereby ensuring uniform coverage of the spatial receptive field. **Stage 2** then acts as the supplementation stage: if budget remains, the next highest-scoring token from each grid cell that has not yet been selected enters a round-robin competition. By this mechanism, the Spatial channel no longer purely chases locally high-scoring regions, but instead tends to supplement tokens that are both semantically complementary to the Focal tokens and more evenly distributed across the spatial extent. If the budget is still not exhausted after the grid rounds, global backfill is performed from the remaining candidates following the same redundancy-aware rules, ultimately yielding the Spatial index set $\mathcal{I}_{\text{spa}}$.

Channel merging and sequence reconstruction. Since the Spatial channel operates exclusively on tokens not selected by the Focal channel, the two index sets are disjoint by construction. The final retained visual token set is:

$$\mathcal{I} = \mathcal{I}_{\text{foc}} \cup \mathcal{I}_{\text{spa}}, \quad |\mathcal{I}| = k, \tag{10}$$

sorted by their original positions, concatenated with the text tokens, and fed into the LLM; the original positional encodings of the retained tokens are preserved to maintain spatial alignment.

3.4 Complete Training Objective

The training objective of GAZE must simultaneously satisfy three design requirements: preserving downstream task performance, bridging the selection discrepancy between training and inference, and converting bounding-box annotations into explicit semantic supervision. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_t \cdot \mathcal{L}_{\text{cons}} + \mathcal{L}_{\text{bbox}}, \tag{11}$$

where $\mathcal{L}_{\text{CE}}$ is the standard autoregressive cross-entropy loss of the language model conditioned on the compressed visual tokens, and $\mathcal{L}_{\text{bbox}}$ denotes the localization-aware ranking loss as previously defined in Section 3.2. As described in Section 3.1, inference employs a discrete Top-K hard selection whose gradient is zero. Following standard practice in differentiable token selection [42], during training we apply a sigmoid softening to the importance scores of each channel, producing continuous soft masks $\tilde{m}_{\text{foc}}, \tilde{m}_{\text{spa}} \in [0,1]^N$, while the hard selection simultaneously generates binary hard masks $m_{\text{foc}}, m_{\text{spa}} \in \{0,1\}^N$. The consistency loss is:

$$\mathcal{L}_{\text{cons}} = \mathrm{BCE}(\tilde{m}_{\text{foc}}, m_{\text{foc}}) + \mathrm{BCE}(\tilde{m}_{\text{spa}}, m_{\text{spa}}), \tag{12}$$

whose regularization weight follows a linear schedule over training:

$$\lambda_t = \lambda_{\text{start}} + (\lambda_{\text{end}} - \lambda_{\text{start}}) \cdot \min\left(\frac{n}{T}, 1\right), \tag{13}$$

where λ_{start} and λ_{end} are the initial and final weights, n is the current training step, and T is the total number of steps. At the beginning of training, λ_t is small, allowing VIS to prioritize establishing foreground–background discriminability before the consistency constraint intervenes, thereby avoiding gradient conflicts between premature binary alignment and the ranking gradients of $\mathcal{L}_{\text{bbox}}$. As training proceeds, λ_t increases linearly to λ_{end}, gradually and progressively hardening the soft masks of both the Focal and Spatial channels toward their respective Top-K binary masks. This eliminates the train–inference discrepancy at test time.

The three loss terms are mutually complementary: $\mathcal{L}_{\text{CE}}$ ensures that the compressed visual representation can still support high-quality generation; $\mathcal{L}_{\text{cons}}$ bridges the distributional gap between training and inference; and $\mathcal{L}_{\text{bbox}}$ provides an explicit spatial prior on semantic importance. The entire model is optimized end-to-end without any additional multi-stage training procedures.

4 Experiments

4.1 Experimental Setup

Models. We evaluate GAZE on two representative vision-language models with diverse architectures and resolution strategies: Qwen2.5-VL-7B-Instruct [3] and LLaVA-OneVision-1.5-8B [19]. All experiments are conducted on a single 24 GB NVIDIA RTX 4090.

Table 1: Performance comparison of visual token pruning methods under three compression ratios on Qwen2.5-VL-7B-Instruct across eight benchmarks spanning document understanding, chart reasoning, OCR, and general VQA. Avg (%) is the mean score normalized by the full-token baseline (100%). * denotes training-based methods. Our GAZE consistently achieves the best trade-off between efficiency and accuracy across all ratios.

Method	DocVQA Anls	ChartQA Relaxed	OCRBench Acc	ScienceQA EM	AI2D EM	MMMU Acc	MME Score	POPE F1	Avg
Dynamic Resolution (MinPix=256×28×28, MaxPix=2048×28×28), Upper Bound (100%)									
Qwen2.5-VL-7B-Instruct	94.33	83.40	83.80	87.26	93.59	50.78	2342.15	86.19	100%
Retain 30% Tokens (70% Compression Ratio)									
FastV [6] (ECCV'24)	84.01	67.64	68.70	83.06	86.92	49.44	2263.58	80.47	90.95
PruMerge+ [31] (ICCV'25)	73.95	74.24	64.80	85.22	82.77	47.67	2239.64	83.69	89.68
DivPrune [1] (CVPR'25)	82.51	67.52	72.00	86.01	88.28	48.33	2224.06	84.68	91.96
PruneSiD [10] (ICLR'26)	76.94	62.52	62.50	83.89	79.92	48.44	2175.74	**86.10**	87.60
V^2Drop [4] (CVPR'26)	79.53	67.88	66.40	**87.16**	**92.78**	50.00	2325.64	86.04	92.70
Dynamic-LLaVA* [13] (ICLR'25)	86.32	68.88	75.00	78.78	83.29	41.78	2180.42	80.87	89.01
VisionSelector* [42] (arXiv'25)	91.59	72.72	79.90	85.77	91.97	**50.11**	2318.27	85.05	96.57
GAZE* (Ours)	**92.97**	**75.92**	**82.00**	86.76	92.65	**50.11**	**2338.60**	85.48	**97.95**
Retain 20% Tokens (80% Compression Ratio)									
FastV [6] (ECCV'24)	75.99	60.48	59.70	82.75	82.35	49.00	2152.74	76.12	85.49
PruMerge+ [31] (ICCV'25)	61.09	52.56	56.20	83.79	77.62	46.11	2219.30	81.74	81.78
DivPrune [1] (CVPR'25)	73.81	57.88	64.80	84.33	82.29	46.33	2198.82	83.55	86.45
PruneSiD [10] (ICLR'26)	67.64	47.48	55.80	83.69	76.45	47.33	2114.91	85.03	81.87
V^2Drop [4] (CVPR'26)	67.94	60.08	57.20	**86.27**	**92.29**	**50.44**	2310.13	**85.54**	88.38
Dynamic-LLaVA* [13] (ICLR'25)	79.21	61.92	67.40	77.00	81.87	42.56	2117.37	77.22	84.77
VisionSelector* [42] (arXiv'25)	88.10	68.20	75.20	85.67	90.15	48.56	2293.54	84.27	93.84
GAZE* (Ours)	**91.00**	**71.12**	**79.20**	84.93	91.87	48.67	**2339.79**	84.53	**95.69**
Retain 10% Tokens (90% Compression Ratio)									
FastV [6] (ECCV'24)	58.64	44.64	44.00	81.95	75.74	45.78	1940.91	65.99	74.08
PruMerge+ [31] (ICCV'25)	42.89	42.84	40.40	82.60	73.09	45.44	1944.04	78.46	72.67
DivPrune [1] (CVPR'25)	54.04	39.80	47.70	82.15	72.80	46.00	2015.66	79.27	75.31
PruneSiD [10] (ICLR'26)	44.52	35.08	39.50	81.11	68.52	45.67	1990.44	80.26	71.33
V^2Drop [4] (CVPR'26)	48.52	47.20	44.30	**85.57**	**91.26**	**49.78**	**2314.49**	**85.31**	81.41
Dynamic-LLaVA* [13] (ICLR'25)	61.17	59.96	55.70	75.71	76.10	43.11	1977.58	70.80	77.85
VisionSelector* [42] (arXiv'25)	75.62	59.80	64.20	83.54	84.00	46.44	2175.75	79.73	86.35
GAZE* (Ours)	**82.34**	**62.04**	**69.10**	82.45	86.37	48.11	2222.20	82.58	**89.54**

Training setup. During training, we freeze the visual encoder, the vision-language projector, and the language model backbone. Only the VIS is optimized. Training data is drawn from the Visual CoT [32] mixture dataset, from which we sample **20,000** examples spanning nine sources: GQA [15], DocVQA [28], TextVQA [33], VSR [22], Visual7W [43], DUDE [34], SROIE [14], Flickr30k [30], and InfographicsVQA [27]. The dataset covers scene-level visual question answering, document OCR, spatial reasoning, and other task types. All samples are annotated with bounding boxes.

Evaluation setup. We conduct experiments across models of different architectures to validate the generalizability of GAZE. We evaluate Qwen2.5-VL-7B-Instruct on eight benchmarks: DocVQA [28], ChartQA [26], OCRBench [24], ScienceQA [25], AI2D [18], MMMU [40], MME [11], and POPE [20], and LLaVA-OneVision-1.5-8B on a subset of five benchmarks, under three token retention ratios: 30%, 20%, and 10%. All experiments use the lmms-eval [41] framework with bfloat16 precision and FlashAttention-2 [7]. Results for additional model variants are provided in the supplementary appendix.

Table 2: Performance comparison of visual token pruning methods on LLaVA-OneVision-1.5-8B. Avg (%) is computed over 5 common benchmarks.

Method	DocVQA Anls	ChartQA Relaxed	OCRBench Acc	AI2D EM	MME Score	Avg
LLaVA-OV-1.5-8B	97.87	86.72	83.00	94.01	2271.32	100%
Retain 30% Tokens (70% Compression Ratio)						
FastV [6] (ECCV'24)	86.51	68.64	59.60	83.52	2019.84	83.42
DivPrune [1] (CVPR'25)	85.23	70.64	64.50	89.35	2104.77	86.79
GAZE (Ours)	**91.43**	**77.60**	**72.40**	**90.42**	**2208.86**	**92.71**
Retain 20% Tokens (80% Compression Ratio)						
FastV [6] (ECCV'24)	76.64	59.08	48.90	78.89	1918.66	74.75
DivPrune [1] (CVPR'25)	75.90	58.64	55.60	84.26	2057.83	78.48
GAZE (Ours)	**85.99**	**71.68**	**68.10**	**89.12**	**2127.62**	**88.21**
Retain 10% Tokens (90% Compression Ratio)						
FastV [6] (ECCV'24)	54.43	38.08	37.40	73.15	1800.00	60.33
DivPrune [1] (CVPR'25)	56.48	38.24	41.90	76.85	1897.29	63.51
GAZE (Ours)	**71.87**	**59.00**	**56.70**	**84.33**	**2039.34**	**77.85**

Table 3: GAZE component ablation on Qwen2.5-VL-7B (10% token retention). The base model uses $\mathcal{L}_{CE}$ loss with Top-K selection; bounding-box supervision, redundancy penalty, and Focal–Spatial dual-channel selection are added incrementally. Avg (%) is the mean score over five benchmarks, normalized against the full GAZE configuration (100%).

Components			Benchmarks					Avg
BBox	Redun.	Foc-Spa	ChartQA	OCRBench	AI2D	MME	POPE	
			46.12	53.90	77.75	1890.15	70.59	82.58
✓			57.32	66.30	84.42	2168.87	79.81	96.07
✓	✓		61.56	68.10	85.59	2201.97	81.51	98.93
✓	✓	✓	**62.04**	**69.10**	**86.37**	**2222.20**	**82.58**	**100.00**

4.2 Main Results

Results on Qwen2.5-VL-7B-Instruct. We compare against five training-free methods: FastV [6], PruMerge+ [31], DivPrune [1], PruneSiD [10], and V^{2}Drop [4], as well as two training-based methods: Dynamic-LLaVA [13] and VisionSelector [42]. As shown in Table 1, GAZE achieves the highest average accuracy retention across all three retention budgets. The margin over competing methods widens consistently as the retention rate decreases. At the highest compression level, with only 10% tokens retained, GAZE surpasses the second-best method, VisionSelector, by 3.19%. The advantage is particularly pronounced on document understanding benchmarks: GAZE leads by 6.72 points on DocVQA, where the localization-aware ranking loss explicitly prioritizes small text regions. At 20% retention, GAZE achieves 2339.79 on MME, nearly matching the full-token baseline of 2342.15. This result indicates that foreground-prioritized token retention preserves global visual perception capability even under high compression ratios.

Results on LLaVA-OneVision-1.5-8B. According to the experimental results summarized in Table 2, our GAZE framework achieves substantial performance gains over all baselines on the LLaVA-OneVision-1.5-8B across all three budgets. Under an extreme compression budget of 10%, the average accuracy retention reaches 77.85%, surpassing the second-best method DivPrune by 14.34% .

Table 4: Inference efficiency on Qwen2.5-VL-7B-Instruct. Under 10% token retention, GAZE achieves the lowest prefill latency and peak GPU memory usage, demonstrating superior hardware efficiency compared to state-of-the-art methods.

Method	Prefill (ms)	Total (ms)	Peak Mem. (GB)	KV Cache (MB)
Original	335.12	486.45	16.22	109.03
PruMerge+ [31]	239.29	372.75	15.93	**12.93**
V^{2}Drop [4]	214.66	361.08	15.80	43.92
GAZE (Ours)	**186.48**	**337.51**	**15.76**	12.94

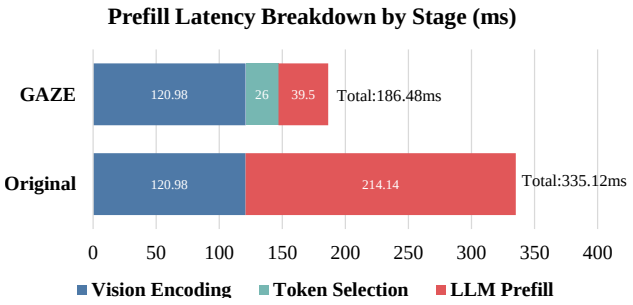


Figure 3: Prefill latency breakdown by stage (ms) on Qwen2.5-VL-7B-Instruct at a 10% budget, measured on a single NVIDIA RTX 4090. Token Selection introduces minimal overhead (~26 ms) while LLM Prefill latency is reduced by 81.5%.

On DocVQA, GAZE scores 71.87 versus DivPrune's 56.48, a margin of 15.39 points. This result highlights the effectiveness of GAZE on text-dense document understanding tasks.

4.3 Ablation Study

Component Ablation. We conduct a stepwise ablation on Qwen2.5-VL-7B-Instruct at a 10% budget, normalizing Avg against the full GAZE configuration. Results are shown in Table 3. Starting from a base model trained with $\mathcal{L}_{CE}$ and standard Top-K selection, adding bounding-box supervision raises Avg from 82.58% to 96.07%, the largest single gain. This improvement comes from two components working in concert. The ranking loss $\mathcal{L}_{bbox}$ injects foreground–background semantic priors into the scorer. The adaptive budget allocation adjusts the retention ratio dynamically according to foreground area. Together, they enable VIS to accurately identify critical regions even under aggressive compression. Further adding the deduplication penalty brings Avg to 98.93%, effectively discouraging the scorer from repeatedly selecting redundant tokens with overlapping semantic content. Incorporating Focal–Spatial dual-channel selection yields consistent gains across all five benchmarks, providing richer scene context for holistic perception tasks.

4.4 Efficiency Analysis

Inference Efficiency. Table 4 reports the efficiency comparison of GAZE against the full-token baseline and representative pruning methods on Qwen2.5-VL-7B-Instruct, evaluated on a single RTX

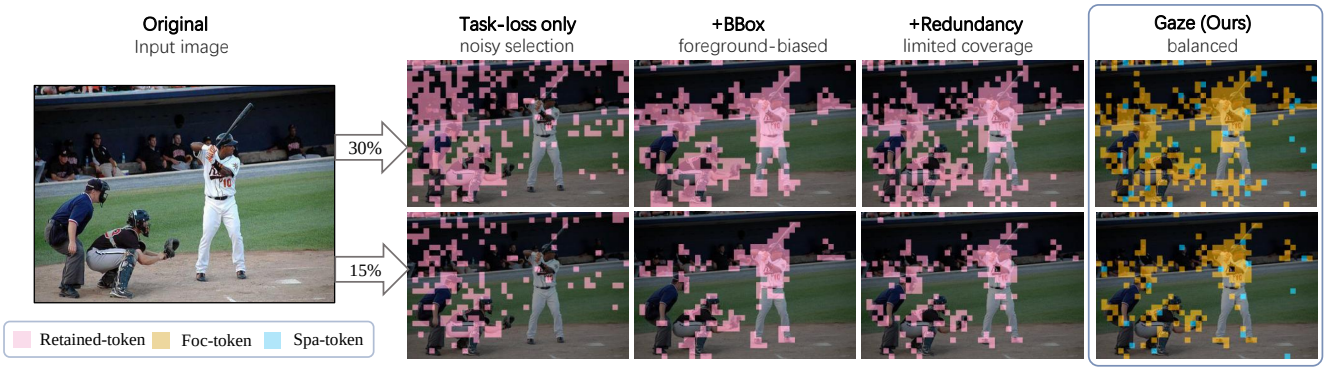


Figure 4: Token selection visualization across ablation variants at retention rates of 30% and 15%, with each variant progressively adding one component (+BBox supervision, +redundancy penalty, +Focal–Spatial selection). The full GAZE model achieves a balanced focal–spatial coverage even at extreme compression.

Table 5: Training efficiency and benchmark performance at 10% token retention on Qwen2.5-VL-7B-Instruct. Data, GPU, and Time denote the number of training samples, the number of RTX 4090 (24GB) GPUs, and wall-clock training duration, respectively. GAZE attains better performance at substantially lower training cost.

Method	Training Cost			Benchmarks			
	Data	GPU	Time	ScienceQA	OCRBench	MME	POPE
Dynamic-LLaVA	665K	2×	~16 h	75.71	55.70	1977.58	70.80
VisionSelector	144K	1×	~5 h	**83.54**	64.20	2175.75	79.73
GAZE	15K	1×	~55 min	82.50	65.80	2149.21	79.74
GAZE	25K	1×	~90 min	83.39	**70.00**	2207.38	**83.97**
GAZE (Ours)	**20K**	**1×**	**~70 min**	82.45	69.10	**2222.20**	82.58

4090 with a batch size of 1. At a 10% token budget, GAZE achieves approximately 1.8× prefill speedup and 8.4× KV-cache compression, while also reducing peak GPU memory and total inference time. GAZE outperforms both PruMerge+ and V2Drop across all four efficiency metrics. Figure 3 breaks down the prefill latency by stage and shows that the token selection module introduces only ~26 ms of overhead. The dominant source of speedup is the substantial reduction in LLM prefill time, confirming that VIS is lightweight and imposes negligible impact on inference throughput.

Training Cost. Compared to existing training-based methods, GAZE requires only 20K training samples and completes training in approximately 70 minutes on a single RTX 4090 (24GB). This corresponds to 3.0% of the data used by Dynamic-LLaVA and 13.9% of that used by VisionSelector, substantially reducing training cost. Further analysis reveals that with only 15K samples, GAZE still outperforms or matches VisionSelector on OCRBench and POPE, despite VisionSelector using 144K samples. This highlights the strong data efficiency of our approach. Increasing the data size to 25K yields modest improvements on some metrics, but extends training time from approximately 70 to 90 minutes with diminishing returns. We therefore adopt 20K as the default configuration, providing a favorable balance between performance and training cost.

4.5 Qualitative Visualization

As shown in Figure 4, we visualize the token selection patterns of different ablation variants at retention rates of 30% and 15%. When trained with task loss only, the selected tokens are scattered across the image without consistent coverage of foreground regions. Adding bounding-box supervision concentrates selection on foreground objects, but leads to excessive token clustering and neglects surrounding spatial context. Incorporating the redundancy penalty yields a more dispersed distribution, yet systematic background coverage remains limited. The full GAZE model exhibits a distinctly different selection pattern under both budgets. Focal tokens concentrate on semantically salient regions, while spatial tokens provide uniform coverage over the remaining area, jointly preserving foreground details and global context. Even at an extreme retention rate of 15%, GAZE maintains coherent modeling of target objects and their scene relationships, whereas all other variants exhibit clear coverage degradation at the same budget.

5 Conclusion

In this paper, we propose GAZE, an end-to-end framework for grounding-aware visual token pruning. GAZE effectively resolves the inherent conflict between semantic discriminativeness and spatial coverage in visual token pruning. GAZE leverages a localization-aware ranking loss, an adaptive budget allocation strategy, and a Focal–Spatial dual-channel selection mechanism equipped with a shared redundancy penalty. Together, these enable the framework to precisely identify token subsets that balance semantic saliency and spatial coverage, without requiring any bounding box annotations during inference. Extensive experimental results demonstrate that GAZE consistently outperforms existing methods across multiple benchmarks, with its performance advantages becoming particularly more pronounced at higher compression ratios. While the current approach relies on bounding box annotations for training, future work could explore leveraging the segmentation capabilities of vision foundation models or self-supervised saliency signals as annotation-free alternatives, to further expand the applicability.

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